

WHERE ARE WE GOING?

UNIT 1

FOUNDATIONAL CHEMISTRY SKILLS

Chapters 1 – 5

Weeks 1 – 4

Subatomic particles
Atoms, molecules, ion
Periodic table of elements
Avogadro's and the mole
Chemical formulas
Chemical equations
Stoichiometry

UNIT 2

HOW REACTIONS HAPPEN

Chapters 7, 14 – 16

Weeks 5 – 7

Gases
Kinetic molecular theory
Kinetics
Reaction mechanisms
Equilibria

UNIT 3

WHAT REACTIONS PROVIDE

Chapters 6, 18, 19

Weeks 8 – 10

Heat and enthalpy
Work
Gibbs free energy
Electrochemistry

UNIT 4

MICROSCOPIC TO MACROSCOPIC

Chapters 8 – 13

Weeks 12 – 16

Quantum model of atom
Orbitals
Electron configurations
Periodic trends
Lewis structures + VSEPR
Bonding theory
Molecular polarity
Intermolecular forces
Properties and solubility

8B

QUANTUM MODEL OF ATOM

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

Electrons as Waves?

Chemistry in Context

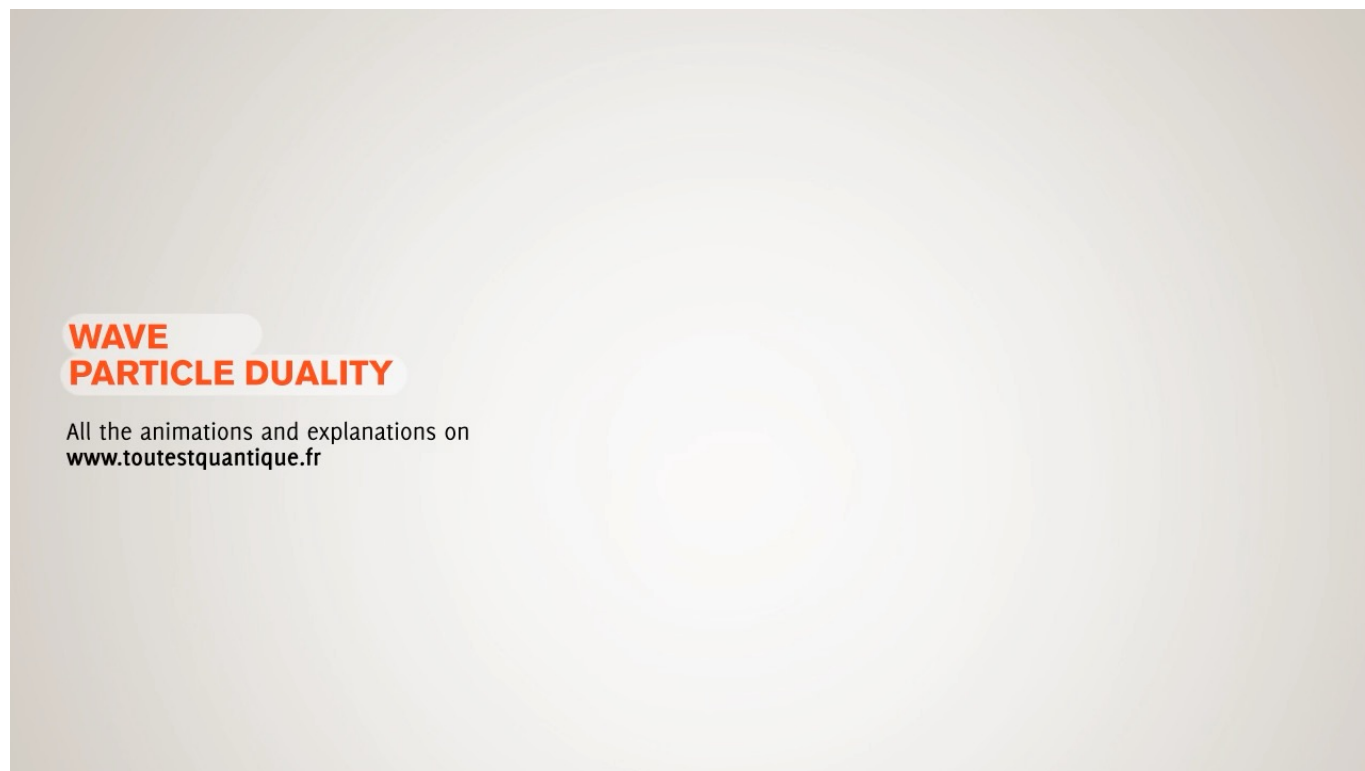
Unfortunately, the Bohr Model fails beyond the hydrogen atom.

- It's *still* a very useful model and we still use it when we can.

Light was treated as both a *wave* and a *particle*.



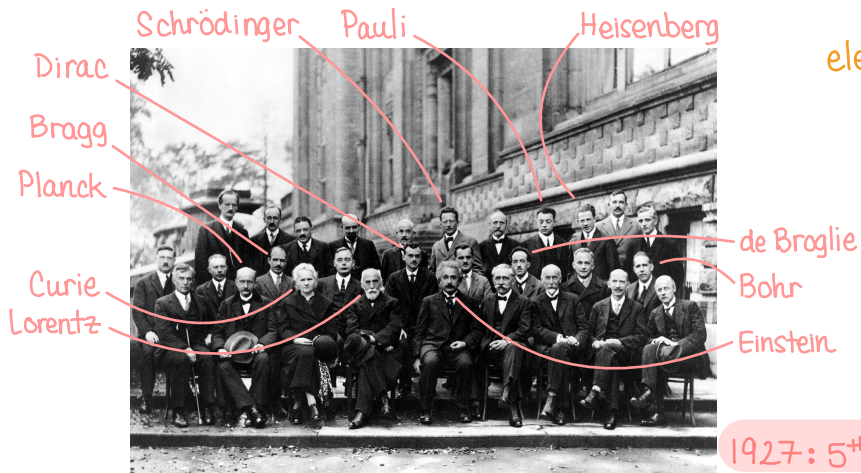
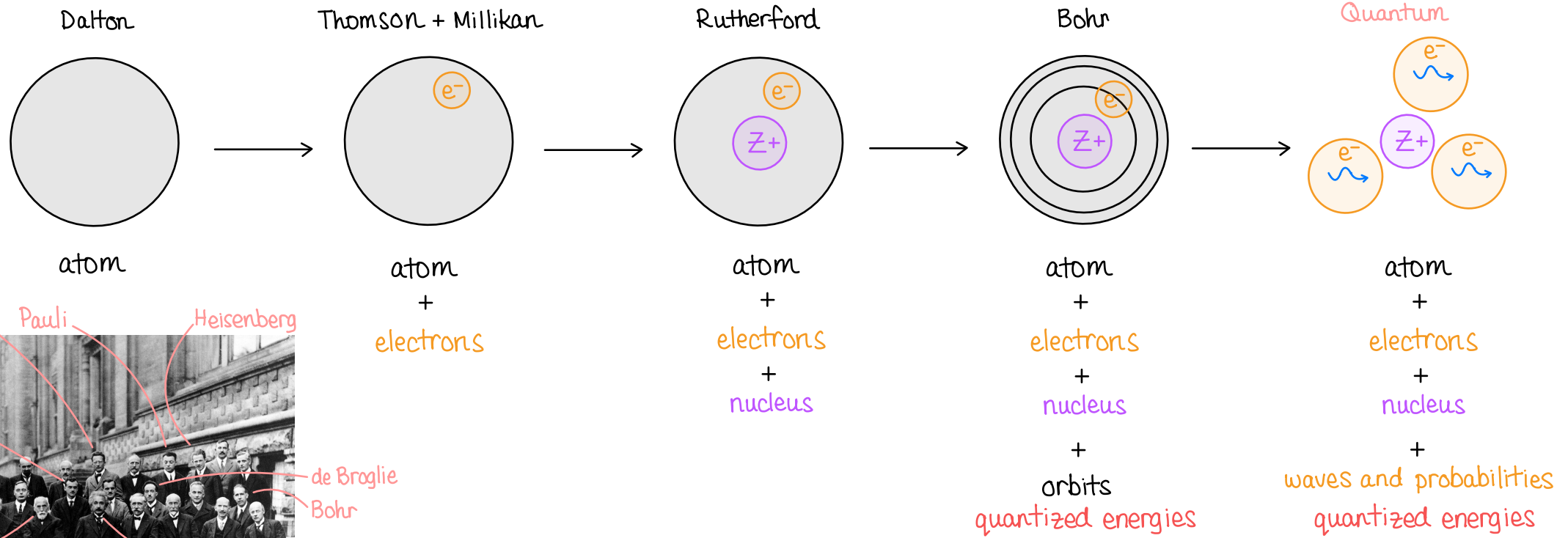
Electrons also need to be treated as both a *particle* and a *wave*.



Electrons as Waves?

Chemistry in Context

The current model of the atom is much more complex, less easy to visualize, and much more theoretical.



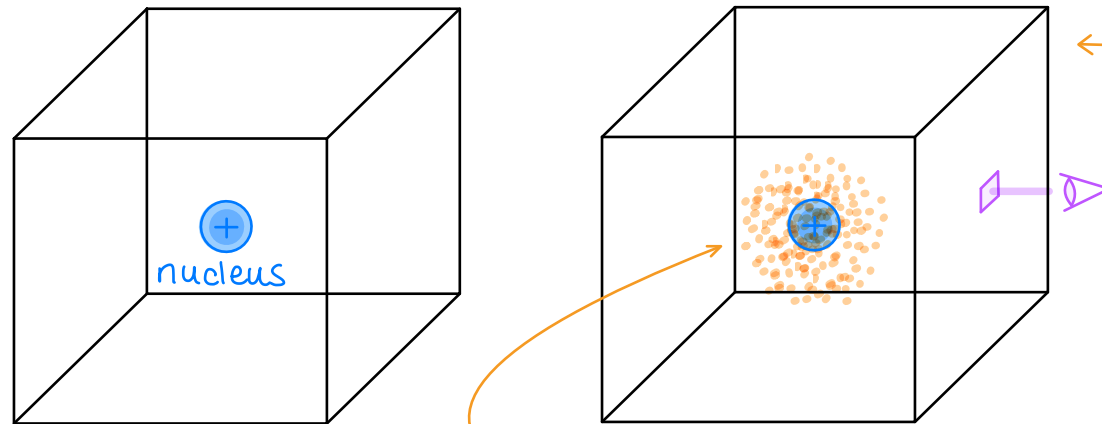
1927: 5th Solvay Conference on "Electrons and Photons"

Orbitals and Quantum Numbers

8.4

Electrons are not confined to circular orbits as postulated by the Bohr model.

- Electrons can be found *nearly anywhere* in the *space* around the nucleus.
- The *probability* of finding an electron in some regions around the *nucleus* is much greater than in other regions.
- The distribution of the electron's probable location is called an ORBITAL.



each dot (•) represents
a probable location of the e^-

Orbitals and Quantum Numbers

8.4

$$\hat{H}\Psi = E\Psi$$

The Schrödinger wave equation can be solved to give 3D distribution maps for the *probable* position/location of an electron in an atom.

- This is the basis of the quantum mechanical model of the atom.

	less accurate but simple	more accurate but complex
	Bohr Model of the Atom	Quantum Model of the Atom
Electrons <i>behave</i> like ...	particles	waves + particles
Electrons <i>exist</i> ...	on 2D circular orbit	in spherical 3D orbitals
The <i>position</i> of an electron is ...	definite	probable
The <i>energy</i> of an electron is ...	quantized/discretized	quantized

Summary *before* we get lost

Chemistry in Context

<u>Energy Level (Shell)</u>	<u>Region/Types (Subshell)</u>	<u>Orbitals and Quantities</u>			
$n=1$	1 region of probability (1 type of orbital)	s (1)			
$n=2$	2 regions of probability (2 types of orbitals)	s (1)	p (3)		orbitals come in different shapes, sizes, and energies !
$n=3$	3 regions of probability (3 types of orbitals)	s (1)	p (3)	d (5)	
$n=4$	4 regions of probability (4 types of orbitals)	s (1)	p (3)	d (5)	f (7)

- Each new shell (energy level n) → one more kind of orbital
- Odd number of orbitals for each subshell (type) → 1, 3, 5, 7, etc.
- Total number of orbitals within a shell (n) → n^2
- 2 electrons can fit in *each* orbital → $2e^-$ in s, $6e^-$ in p, $10e^-$ in d, $14e^-$ in f, etc.

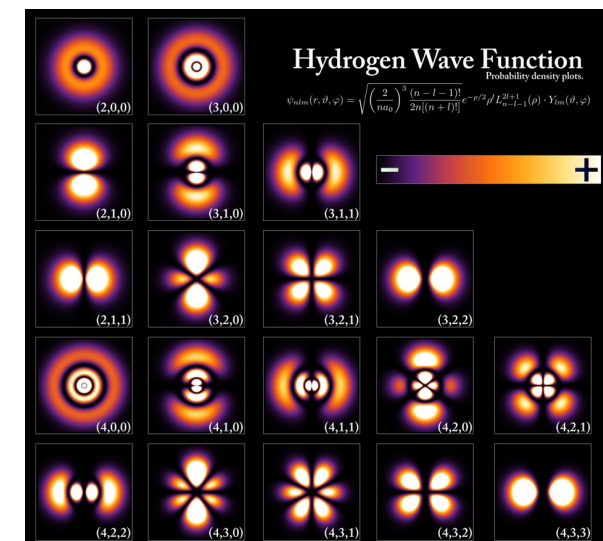
Orbitals and Quantum Numbers

8.4

Quantum numbers

- The names attached to atomic orbitals that come from the functions that solve the wave equations. $\hat{H}\Psi = E\Psi$
- The solutions to Schrödinger's equation: wavefunction: $\Psi(n, \ell, m_\ell)$

each orbital has unique (n, ℓ, m_ℓ) !
 each electron has unique (n, ℓ, m_ℓ, m_s) !



Quantum Number	Symbol	Allowed Values	Meaning
Principal	n	1, 2, 3, ...	Energy level and size of orbital
Angular momentum	ℓ	0, 1, 2, 3, ... $(n-1)$	Shape of orbital
Magnetic	m_ℓ	$-\ell, \dots, 0, \dots, +\ell$	Orientation of orbital; # of orbitals
Spin	m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	Orientation of electron

Orbitals and Quantum Numbers

8.4

Principal quantum number (n)

- Also called: energy level, **shell**, or principal energy level
- Describes the main energy level (shell) and size of an orbital
 - Very similar to n in the Bohr Model!
- May take on any positive integer value: **$n = 1, 2, 3, \dots$**
- Larger values of n indicate:
higher energy orbital
larger (size) orbital
farther from nucleus
similar to n in Bohr model

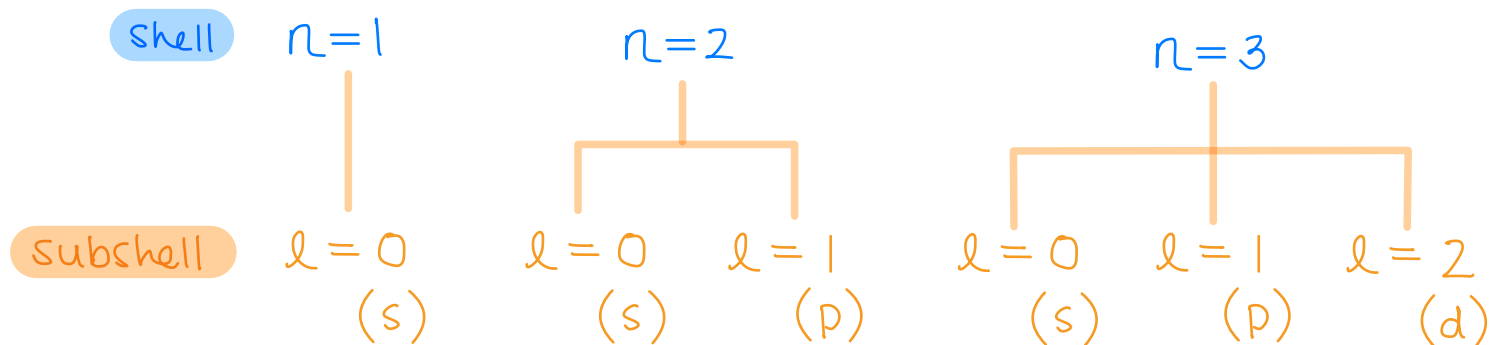
Orbitals and Quantum Numbers

8.4

Angular momentum quantum number (ℓ)

- Also called: **subshell** or sublevel of an orbital
- Describes the overall shape of an orbital
- Allowed values of ℓ depend on n : $\ell = 0, 1, 2, 3, \dots (n-1)$
- Each ℓ is also given a unique subshell letter

Value of ℓ	Subshell
0	s
1	p
2	d
3	f
4 and beyond	alphabetical

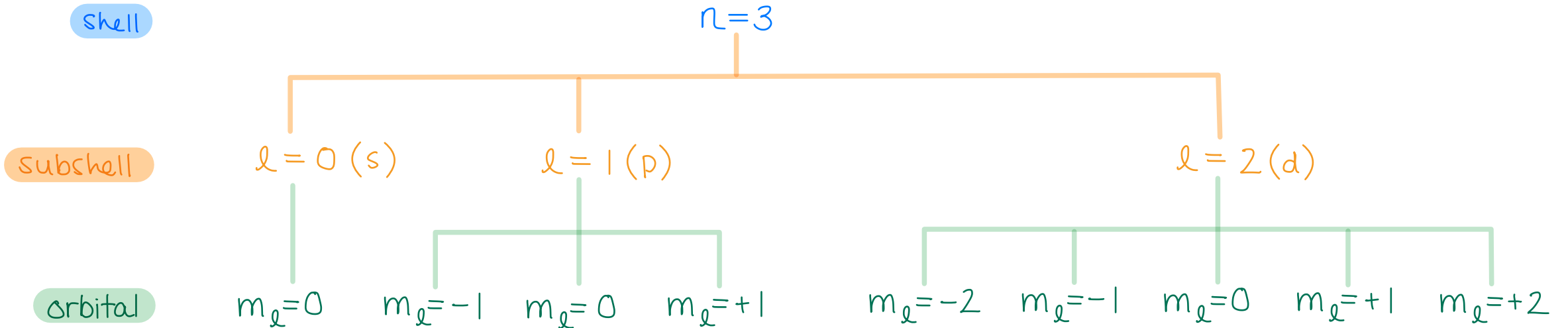


Orbitals and Quantum Numbers

8.4

Magnetic quantum number (m_ℓ)

- Describes the **orientation** in 3D-space (relative to x, y, z axes) of an orbital
- Allowed values of m_ℓ depend on ℓ : $m_\ell = -\ell, \dots, 0, \dots, +\ell$



Orbitals and Quantum Numbers

8.4

Magnetic quantum number (m_ℓ)

- The number of possible values of m_ℓ is equal to the number of orbitals in a set of that type of orbitals.

e.g.) $\ell = 3 \rightarrow m_\ell = \{-3, -2, -1, 0, +1, +2, +3\} \rightarrow 7 \text{ } m_\ell \text{ values} \rightarrow 7 \text{ f orbitals}$
(f)

Value of ℓ	Subshell	Number of orbitals in subshell	Maximum number of electrons in subshell
0	s	1	2
1	p	3	6
2	d	5	10
3	f	7	14
4 and beyond	alphabetical	$2\ell + 1$	2e ⁻ per orbital

2e⁻ per orbital

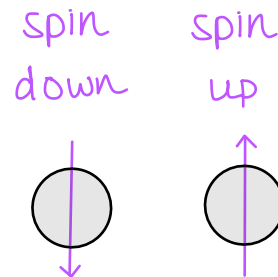
2e⁻ per orbital

Orbitals and Quantum Numbers

8.4

Spin quantum number (m_s)

- Describes the “spin” orientation of an electron in an orbital
- Electrons don't really “spin”
→ just have an “intrinsic” angular momentum
- Electron spin is a fundamental property of an electron (like its charge)
- Only two allowed values of m_s : $m_s = -\frac{1}{2}$ or $+\frac{1}{2}$

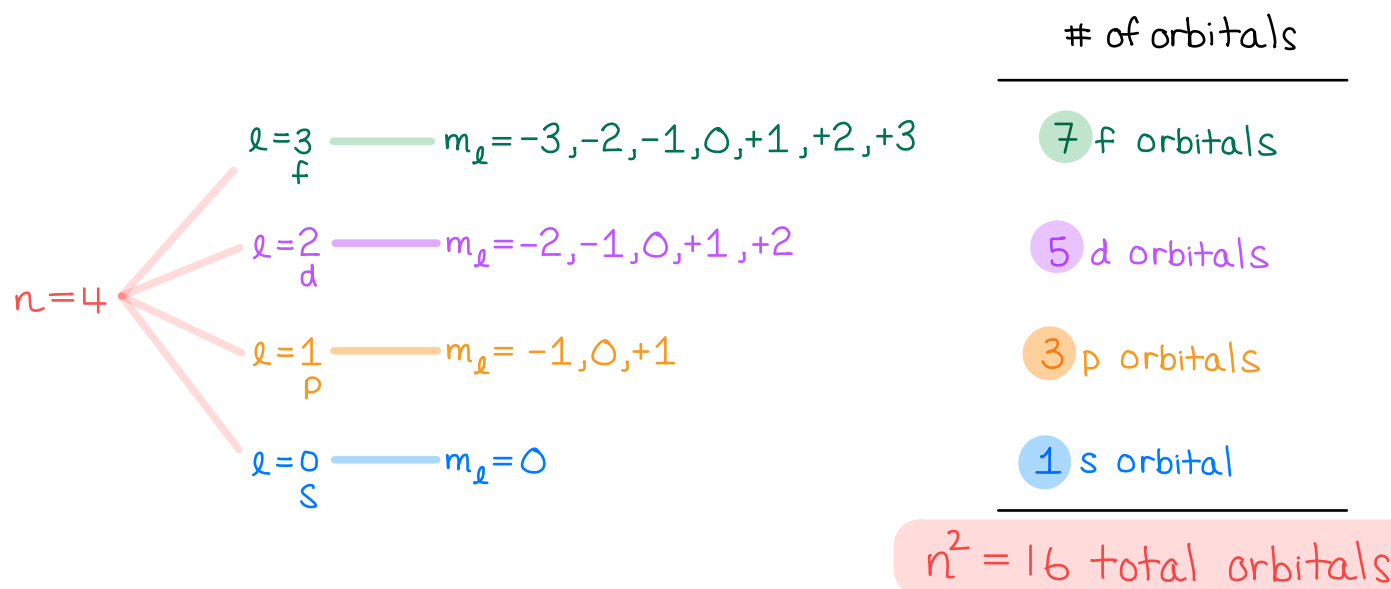


EXAMPLE PROBLEM

Learning Objective(s): 8.5, 8.6

How many total orbitals exist in a shell with principal quantum number $n = 4$?

Consider all subshells!



IN-CLASS QUESTION 1

Learning Objective(s): 8.5, 8.6

What type of orbital is described by the following sets of quantum numbers.

Give your answer using the orbital shorthand notation: (shell)(subshell)

	Set of Quantum Numbers	Orbital
(a)	$n = 1, \ell = 0, m_{\ell} = 0$	1s
(b)	$n = 3, \ell = 2, m_{\ell} = +2$	3d
(c)	$n = 4, \ell = 1, m_{\ell} = -1$	4p
(d)	$n = 4, \ell = 0, m_{\ell} = 0$	4s

Shapes of Orbitals

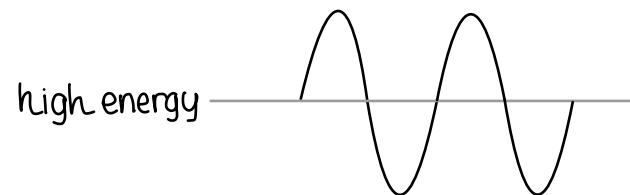
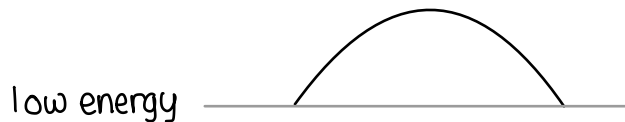
8.5

Every orbital is distinct!

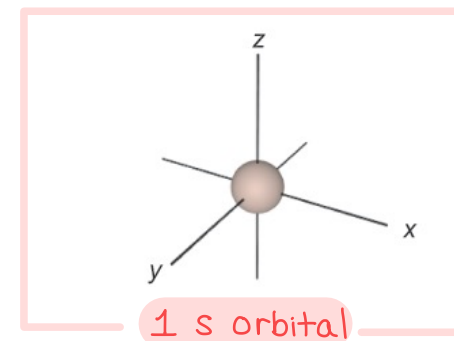
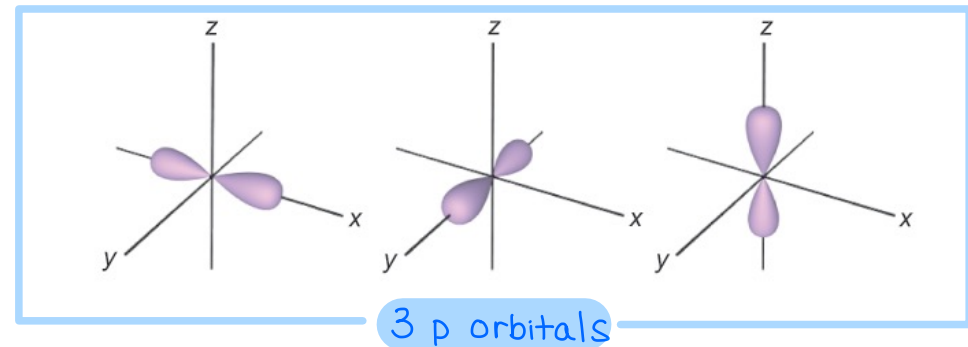
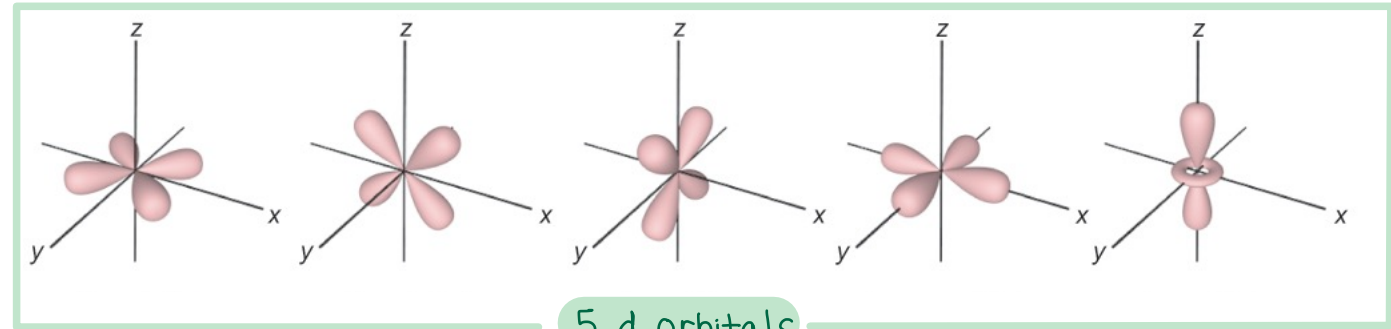
Orbitals represent 3D maps of the probability of finding an electron around the nucleus.

The **s**, **p**, **d**, **f**, etc. orbitals have characteristic *shapes*!

higher energy orbitals are more complex
because higher energy waves are more complex



energy ↑



Shapes of Orbitals

8.5

The ($\ell = 0$) s subshell has one s -orbital.

An s -orbital is *spherically symmetrical* so there is an equal probability of finding an s -orbital electron in any direction around the nucleus.

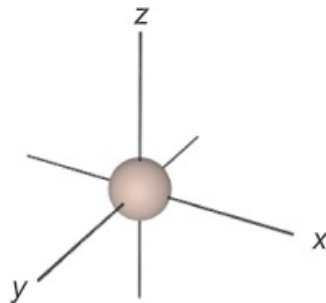


FIGURE 8.17 Shape of s -orbital

The ($\ell = 1$) p subshell has three p -orbitals.

Each p -orbital has two 3D lobes centered on one of the coordinate axes.

These are the three orientations (the three m_ℓ values).

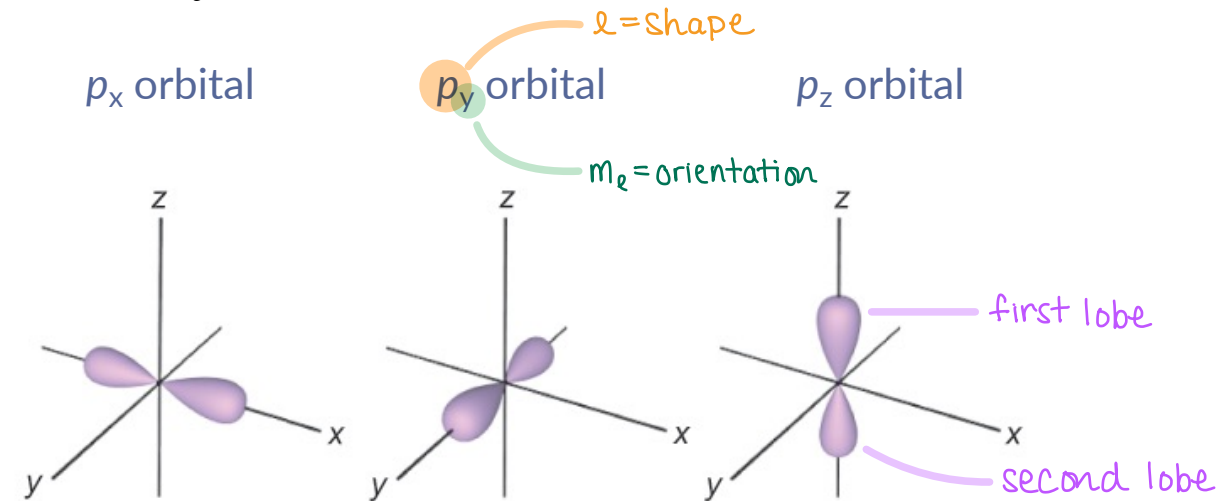


FIGURE 8.17 Shapes of p -orbitals

Shapes of Orbitals

8.5

The ($\ell = 2$) d subshell consists of five d -orbitals.

- Four of the d -orbitals have four 3D lobes in a “cloverleaf” appearance
- The fifth d -orbital is a two-lobed region that lies along the z -axis, with a torus (or ring) horizontally around its center.
- All d -orbitals have two nodes.

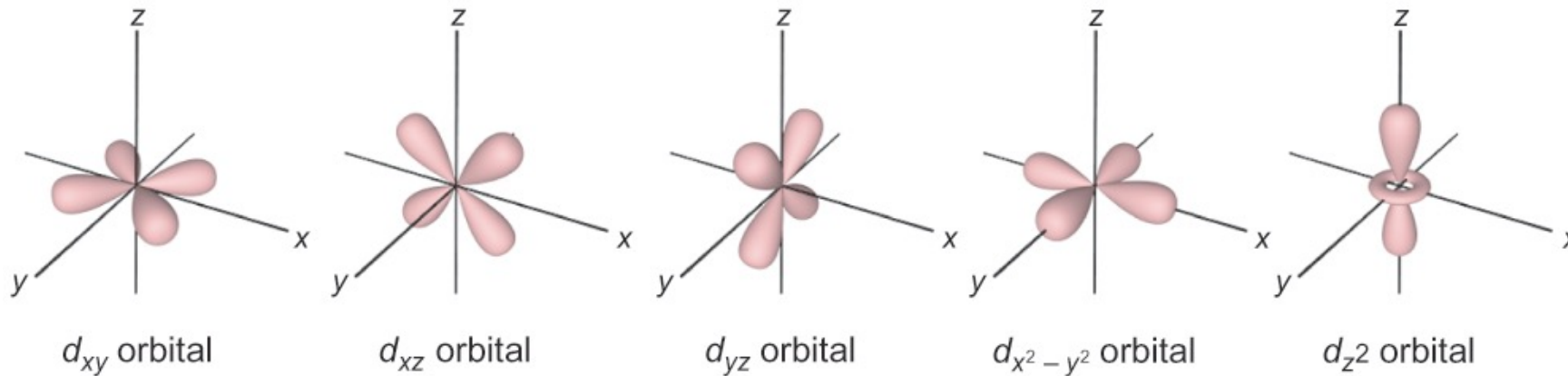


FIGURE 8.17 Shapes of d -orbitals

Shapes of Orbitals

8.5

Relative sizes and energies of orbitals:

- Size and energy generally increase with increasing shell (n)
- Within a shell (n), the energy of subshells is: $E_s < E_p < E_d < E_f$

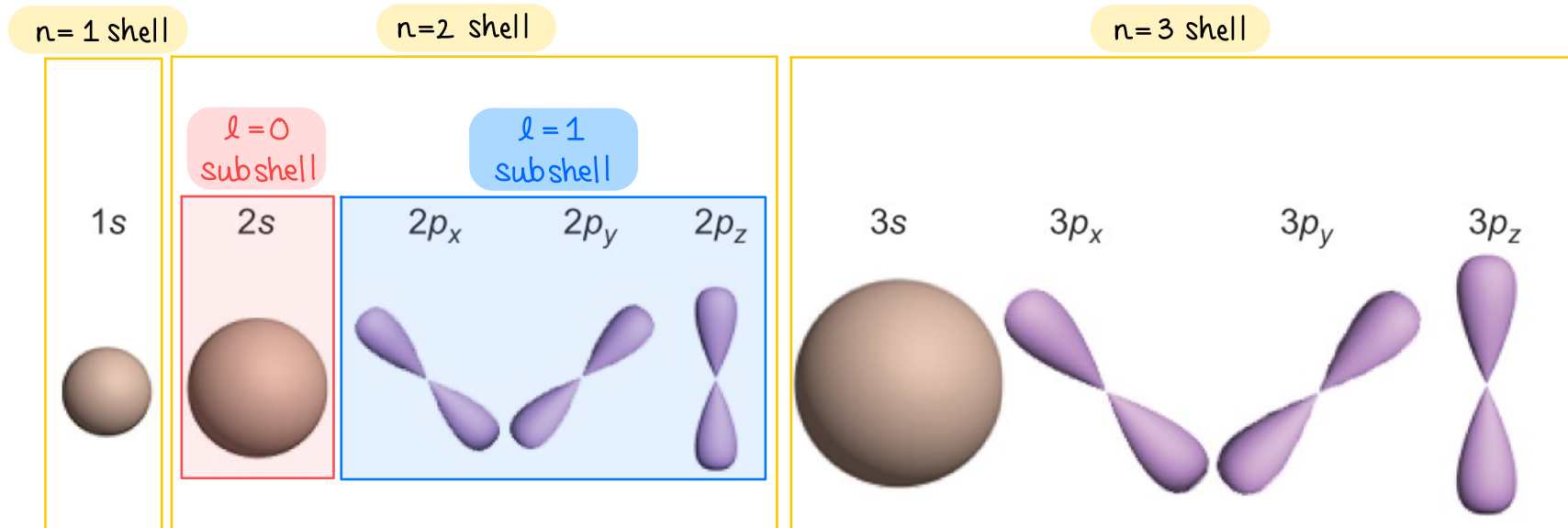


FIGURE 8.18

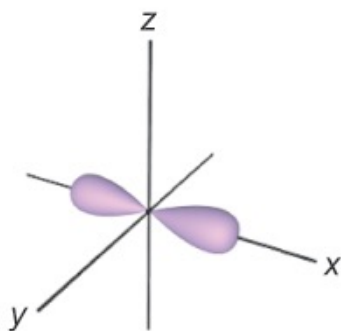
IN-CLASS QUESTION 2

Learning Objective(s): 8.5, 8.6

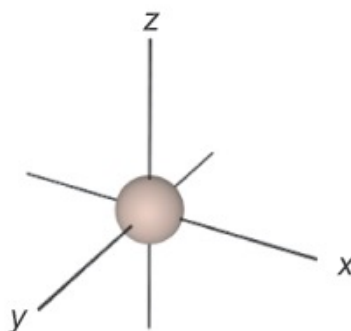
Arrange the orbitals in order of increasing energy within a given shell.

within a shell n :

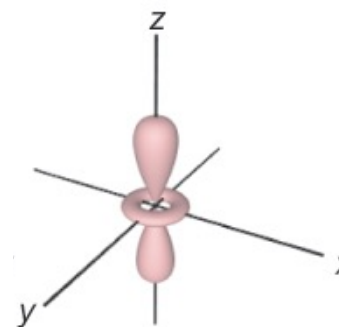
$$E_s < E_p < E_d < E_f$$



Orbital A
p orbital



Orbital B
s orbital



Orbital C
d orbital

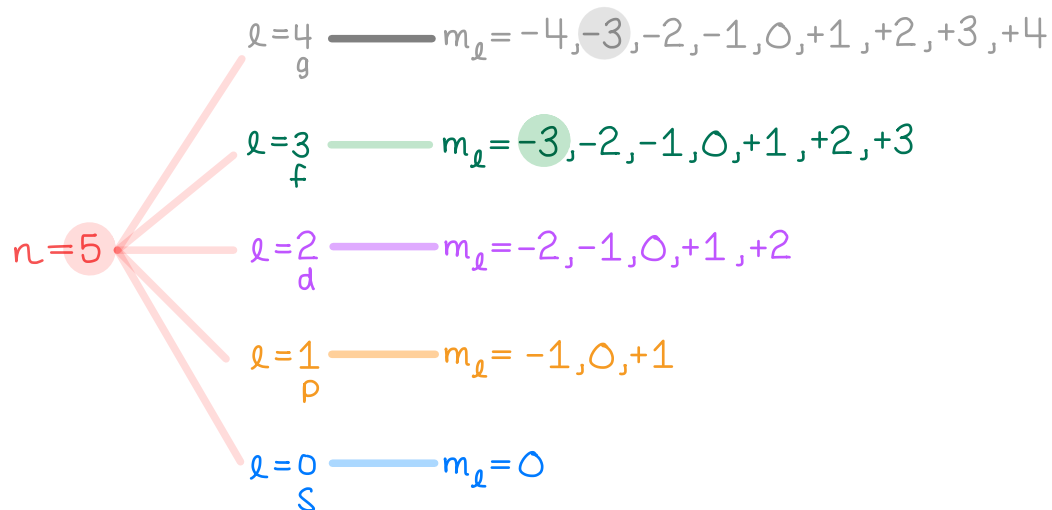
$$B < A < C$$

EXAMPLE PROBLEM

Learning Objective(s): 8.5, 8.6

In an atom, how many orbitals can have the combination of quantum numbers $n = 5$ and $m_\ell = -3$?

Consider all subshells!



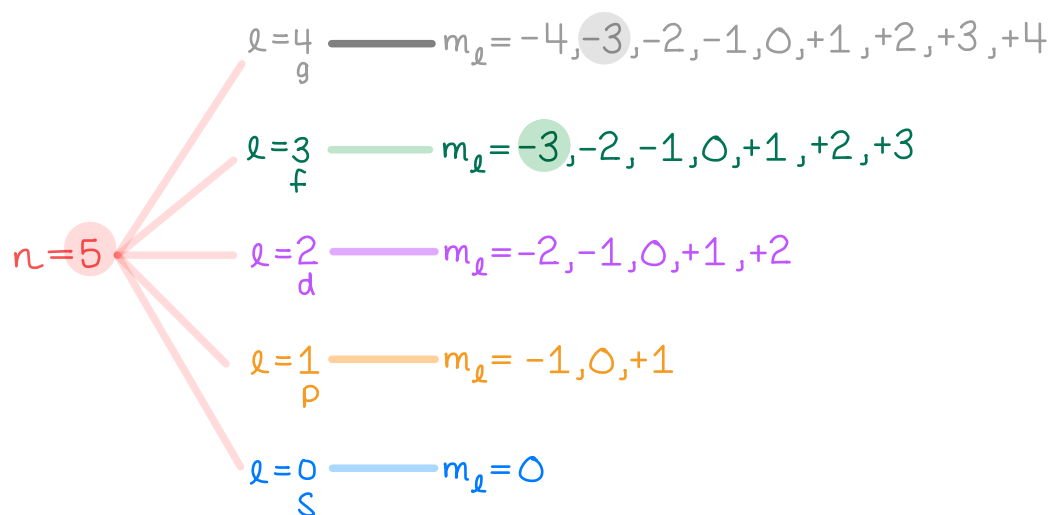
2 orbitals : $(5, 3, -3) = 5f$

$(5, 4, -3) = 5g$

IN-CLASS QUESTION 3

Learning Objective(s): 8.5, 8.6

In an atom, how many *electrons* can have the combination of quantum numbers $n = 5$ and $m_\ell = -3$?



2 orbitals : $(5, 3, -3) = 5f$

$(5, 4, -3) = 5g$

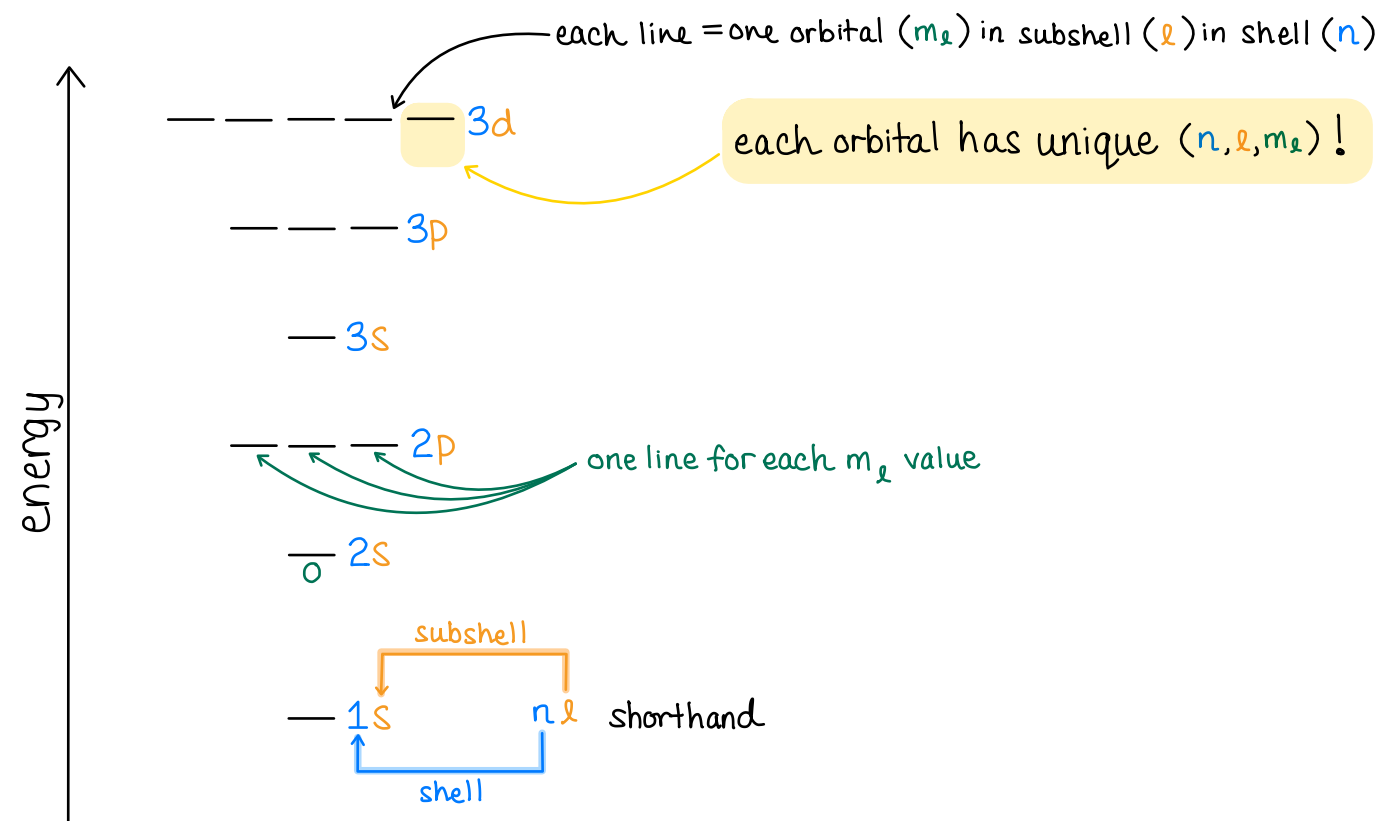
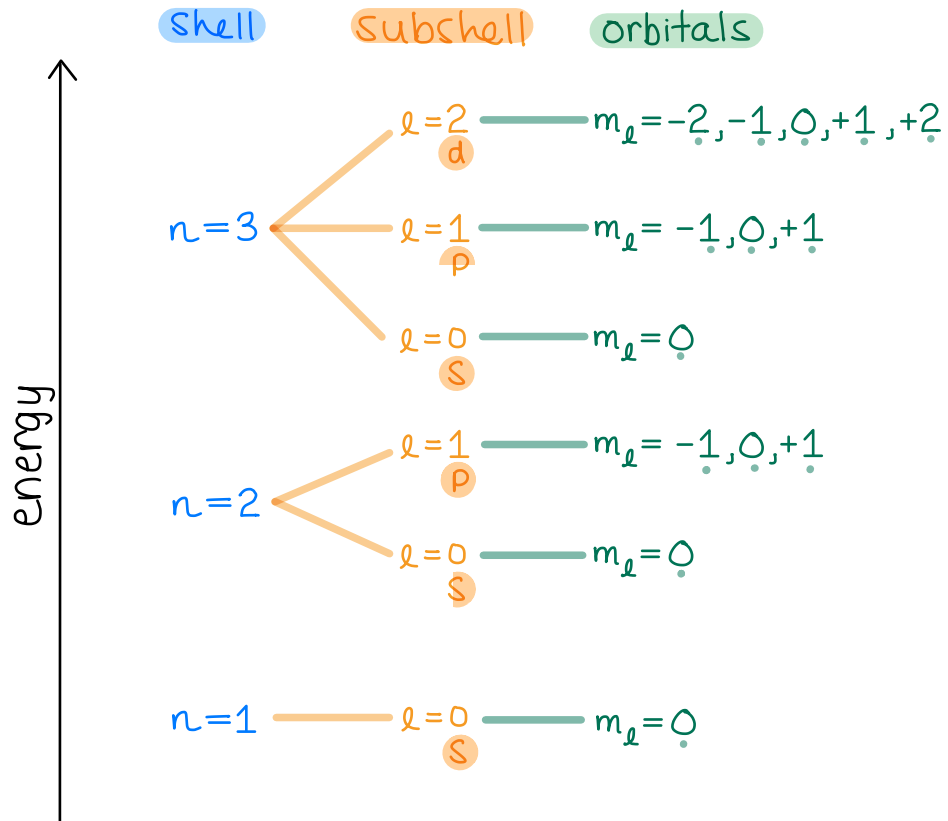
2 e^- in each orbital

4 e^- total

Putting it all together

Chemistry in Context

No two orbitals and no two electrons have the same set of quantum numbers.



BONUS QUESTION

Assessed on: Fall 2021, Exam 2

Arrange the orbitals (defined below) in order of *increasing energy*.

_____ < _____ < _____ < _____

Orbital	Set of Quantum Numbers
Orbital A	$n = 3, \ell = 2, m_\ell = +2$
Orbital B	$n = 4, \ell = 0, m_\ell = 0$
Orbital C	$n = 4, \ell = 2, m_\ell = +2$
Orbital D	$n = 4, \ell = 1, m_\ell = -1$